

SQL QUERY IMPLEMENTATION FOR SOFTWARE MANAGEMENT SYSTEM

```
CREATE DATABASE Software_Management_System;
```

```
USE Software_Management_System;
```

```
SHOW DATABASES;
```

```
CREATE TABLE Department (  
    Department_ID INT PRIMARY KEY,  
    Department_Name VARCHAR(100) NOT NULL,  
    Location VARCHAR(100),  
    Manager VARCHAR(100)  
);
```

```
CREATE TABLE Users (  
    User_ID INT PRIMARY KEY,  
    User_Name VARCHAR(100) NOT NULL,  
    Email VARCHAR(100),  
    Role VARCHAR(50),  
    Department_ID INT,  
    FOREIGN KEY (Department_ID)  
        REFERENCES Department(Department_ID)  
);
```

```
CREATE TABLE Software (  
    Software_ID INT PRIMARY KEY,  
    Software_Name VARCHAR(100) NOT NULL,  
    Version VARCHAR(50),  
    Vendor VARCHAR(100),  
    Category VARCHAR(50),  
    Department_ID INT,  
    FOREIGN KEY (Department_ID)  
        REFERENCES Department(Department_ID)  
);
```

```
CREATE TABLE License (  
    License_ID INT PRIMARY KEY,  
    Software_ID INT,  
    License_Type VARCHAR(50),  
    Expiry_Date DATE,  
    Quantity INT,  
    FOREIGN KEY (Software_ID)  
        REFERENCES Software(Software_ID)  
);
```

```
CREATE TABLE Installation (  
    Installation_ID INT PRIMARY KEY,  
    Software_ID INT,  
    User_ID INT,  
    Install_Date DATE,
```

```

Status VARCHAR(50),
FOREIGN KEY (Software_ID)
REFERENCES Software(Software_ID),
FOREIGN KEY (User_ID)
REFERENCES Users(User_ID)
);

CREATE TABLE Maintenance (
Maintenance_ID INT PRIMARY KEY,
Software_ID INT,
Maintenance_Date DATE,
Type VARCHAR(50),
Remarks VARCHAR(255),
FOREIGN KEY (Software_ID)
REFERENCES Software(Software_ID)
);

SHOW TABLES;

INSERT INTO Department
(Department_ID, Department_Name, Location, Manager)
VALUES
(101, 'Computer Science', 'Block A', 'Dr. Ravi Kumar'),
(102, 'Artificial Intelligence', 'Block B', 'Dr. Priya Sharma'),
(103, 'Electronics', 'Block C', 'Dr. Suresh Rao'),
(104, 'Administration', 'Block D', 'Ms. Anitha');

INSERT INTO Users
(User_ID, User_Name, Email, Role, Department_ID)
VALUES
(201, 'Bhavya', 'bhavya@gmail.com', 'Student', 101),
(202, 'Srujana', 'srujana@gmail.com', 'Faculty', 102),
(203, 'Pravallika', 'pravallika@gmail.com', 'Student', 101),
(204, 'Venkata Raju', 'raju@gmail.com', 'Staff', 103),
(205, 'Tulasi', 'tulasi@gmail.com', 'Faculty', 102);

INSERT INTO Software
(Software_ID, Software_Name, Version, Vendor, Category, Department_ID)
VALUES
(301, 'Microsoft Office', '2021', 'Microsoft', 'Productivity', 101),
(302, 'Python', '3.12', 'Python Software Foundation', 'Programming', 102),
(303, 'MySQL', '8.0', 'Oracle', 'Database', 101),
(304, 'MATLAB', '2024', 'MathWorks', 'Engineering', 103),
(305, 'Visual Studio Code', '1.95', 'Microsoft', 'Development', 102);

INSERT INTO License
(License_ID, Software_ID, License_Type, Expiry_Date, Quantity)
VALUES
(401, 301, 'Commercial', '2027-12-31', 50),
(402, 302, 'Open Source', '2030-12-31', 100),
(403, 303, 'Open Source', '2028-06-30', 40),
(404, 304, 'Academic', '2026-12-31', 30),
(405, 305, 'Free', '2027-06-30', 75);

INSERT INTO Installation
(Installation_ID, Software_ID, User_ID, Install_Date, Status)
VALUES
(501, 301, 201, '2026-01-10', 'Active'),
(502, 302, 201, '2026-01-12', 'Active'),
(503, 303, 203, '2026-02-05', 'Active'),
(504, 304, 204, '2026-02-15', 'Active'),
(505, 305, 202, '2026-03-01', 'Active');

```

```
INSERT INTO Maintenance
(Maintenance_ID, Software_ID, Maintenance_Date, Type, Remarks)
VALUES
(601, 301, '2026-03-10', 'Update', 'Latest security updates installed'),
(602, 303, '2026-03-15', 'Backup', 'Database backup completed'),
(603, 304, '2026-04-05', 'Repair', 'License configuration repaired'),
(604, 302, '2026-04-10', 'Update', 'Python packages updated'),
(605, 305, '2026-04-20', 'Update', 'VS Code updated successfully');
```

```
SELECT * FROM Department;
```

```
SELECT * FROM Users;
```

```
SELECT * FROM Software;
```

```
SELECT * FROM License;
```

```
SELECT * FROM Installation;
```

```
SELECT * FROM Maintenance;
```

```
SELECT
    Software_ID,
    Software_Name,
    Version,
    Vendor,
    Category
```

```
FROM Software;
```

```
SELECT
    S.Software_ID,
    S.Software_Name,
    S.Version,
    S.Vendor,
    D.Department_Name,
    L.License_Type,
    L.Expiry_Date,
    L.Quantity
```

```
FROM Software S
```

```
JOIN Department D
```

```
    ON S.Department_ID = D.Department_ID
```

```
JOIN License L
```

```
    ON S.Software_ID = L.Software_ID;
```

```
SELECT
    L.License_ID,
    S.Software_Name,
    L.License_Type,
    L.Expiry_Date,
    L.Quantity
```

```
FROM License L
```

```
JOIN Software S
```

```
    ON L.Software_ID = S.Software_ID
```

```
ORDER BY L.Expiry_Date;
```

```
CREATE VIEW Software_Installation_View AS
```

```
SELECT
    I.Installation_ID,
    S.Software_Name,
    S.Version,
    U.User_Name,
```

```
U.Role,  
I.Install_Date,  
I.Status
```

```
FROM Installation I  
JOIN Software S  
  ON I.Software_ID = S.Software_ID  
JOIN Users U  
  ON I.User_ID = U.User_ID;  
  
SELECT * FROM Software_Installation_View;
```